Baby Products E-Commerce:

Sales Performance & Customer Behavior Analysis

CDS6324 Data Visualization - Final Project

Term: 2510

Table of Contents

1.0 Business Problem Statament 3

2.0 Project Overview 3

3.0 Data Source 4

4.0 Exploratory Data Analyst (EDA) 5

5.0 Visualizations 6

6.0 Interactivity Features 12

7.0 Design Principles - Shaffer's 4Cs 14

1.0 Business Problem Statement

A Kenya-based baby products e-commerce retailer has accumulated three years of sales transactions, but the management team lacks comprehensive insights into which products, categories, and strategies are truly driving business performance. Without data-driven understanding of customer shopping behaviors, pricing effectiveness, product performance metrics, and channel dynamics, the company risks making suboptimal decisions regarding inventory management, promotional strategies, and resource allocation. There is a critical need to understand sales trends, identify high-value products and customer segments, evaluate the effectiveness of discount strategies, and assess operational efficiency across sales channels.

Question:

How can the business leverage existing sales data uncover actionable insights that will optimize product offerings, pricing strategies, understand customer behavior, and improve operational decisions to sustain growth and profitability?

2.0 Project Overview

This project analyzes 28,755 transaction records from November 2020 to October 2023 to uncover actionable insights across four key areas overall sales performance and trends, product and category revenue contribution, customer shopping behavior by time and channel, and the impact of returns and discount strategies. Nine visualizations were developed across two interactive dashboards built with D3.js to directly address these business concerns and support more informed decisions around inventory, promotions, and channel strategy.

3.0 Data Source

Dataset Name: Ecommerce sales data

Details: This dataset contains real-world retail transaction records collected over a three-year period (November 2020 to October 2023) from a registered e-commerce business based in Kenya, specializing in the sale of baby products and related items.

Number of records/rows: 28755

Number of attributes/columns: 17

Source: Kaggle (Link)

Attributes:

Attributes	Description
Order ID	A unique identifier assigned to each customer order for tracking and reference purposes.
Sales ID	An identifier for the specific sale transaction, linking it to other relevant records and details.
Date	The date on which the sale transaction or order was processed or completed.
Order	Details related to the customer's order linked to a Salesperson for commission processing
Transaction Type	Specifies the nature of the transaction, such as product purchase or shipping.
Sale type	Indicates the type of sale, which might include purchase or return.
Sales channel	Specifies the channel through which the sale was conducted, whether in-store or online.
Product	Identifies the specific product(s).
Net quantity	Represents the quantity of products sold, excluding any returns or discount.
Gross sales	The total revenue generated from sales before accounting for discounts, returns, or other adjustment.
Discounts	The total value or percentage of discounts applied to the sales transactions.
Returns	Represents any products that customers returned, including details such as quantity and value.
Net sales	The total revenue generated from sales after accounting for discounts and returns.
Shipping	The cost associated with shipping the products to the customer, if applicable.
Taxes	Total taxes incurred as part of the sales transaction (noted as 0 since the sales are tax-inclusive, i.e., VAT included)
Total sales	The overall revenue for the sale, typically equivalent to Gross Sales, Net Sales, and Total Sales in this dataset due to tax-inclusivity.

In addition to the original dataset fields, several new columns were derived to enrich the temporal and discount analysis capabilities of the data. These columns are outlined below:

Attributes	Description
Year	Calendar year of the transaction.
Month	Numeric month (1–12).
Month_Name	Full month name (e.g., January, February).
Month_Year	Monthly date used to group transactions by month for trend analysis (standardized to first day of each month).
Week	Day of the month (1–31).
WeekDay	Day of the week (e.g., Monday, Tuesday).
Hour	Hour of the day (0–23)
Discount Rate	The ratio of the discount amount to gross sales for transactions with non-zero discounts. For transactions without discounts, the rate is set to 0.0.
Product Category	A standardized grouping of product types into broader categories to reduce complexity.

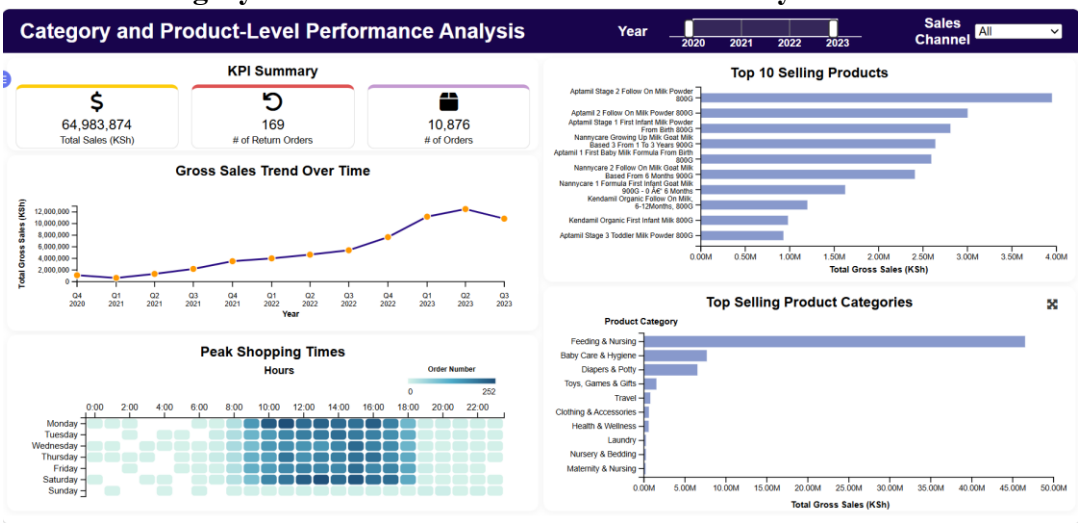
4.0 Exploratory Data Analyst (EDA)

We conducted data preprocessing through the following steps:

- **Data Understanding:**
  used `df.info()` and `df.describe()` to examine structure and statistics.
- **Missing Data Handling:**
  Removed rows with null values in *Product type* and *Product* columns, as mostly are logistics-related entries which are not relevant to our analysis.
- **Handling Inconsistent Data Entries:**
  standardized categorical data by stripping whitespace and converting text to title case for consistency
- **Feature Engineering:**
  - Extracted time features (*year*, *month*, *weekday*, *week*, *hour*) from *Date* column for temporal analysis.
  - **Q4 2023 was excluded** as the dataset only contains data up to October 2023, ensuring consistency in time-based analysis.
  - Created *Discount Rate* (discount/gross sales) with zero-value handling.
  - Grouped *Product type* into broader categories for clearer analysis.

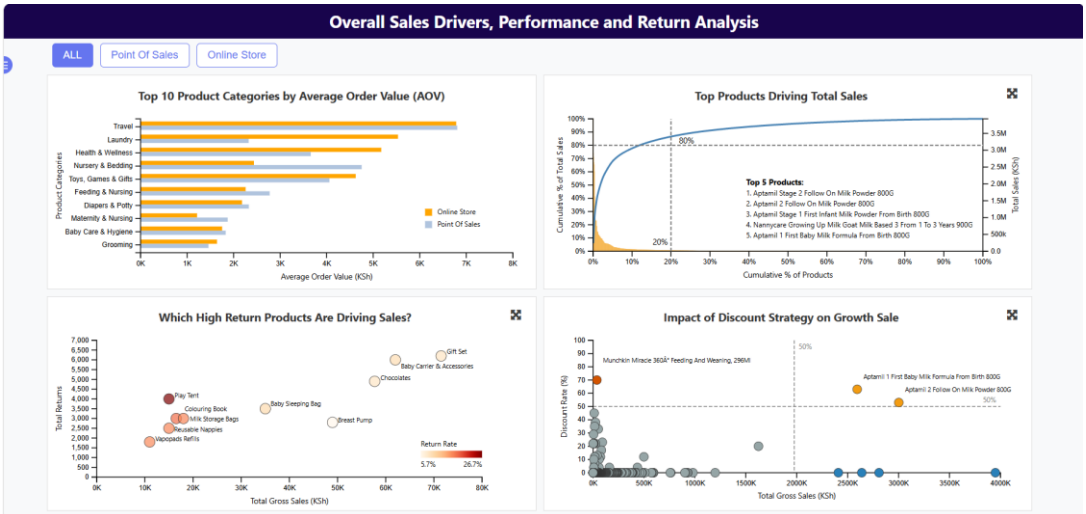
5.0 Visualizations

Dashboard 1: **Category and Product-Level Performance Analysis**



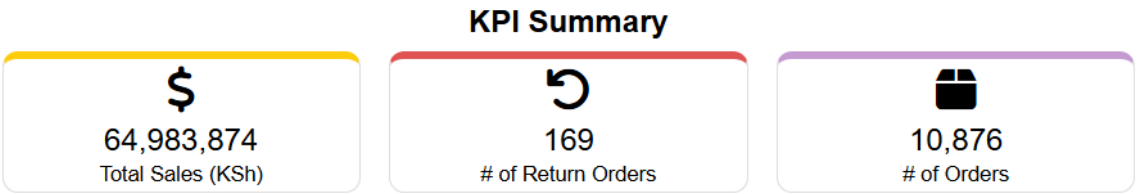
This dashboard is designed to provide a view of sales performance at both the category and product levels. It helps businesses track key performance indicators such as total sales, returns, and order volume over time. By visualizing gross sales trends, top-selling products and categories, and customer shopping patterns by time of day and week, the dashboard supports better decision-making around inventory planning, marketing, and restocking schedules.

Dashboard 2: **Overall Sales Drivers, Performance and Return Analysis**



The purpose is to give viewer a quick view and understanding of where the business is making money, which products and categories need more attention, how returns affect revenues, and whether the discount strategies are effective. By visualizing these metrics, business can identify opportunities to adjust pricing, refine promotions, and optimize inventory. It also helps businesses prioritize high-impact products, focus on top-performing or problematic items, and support decisions that improve profitability and customer satisfaction.

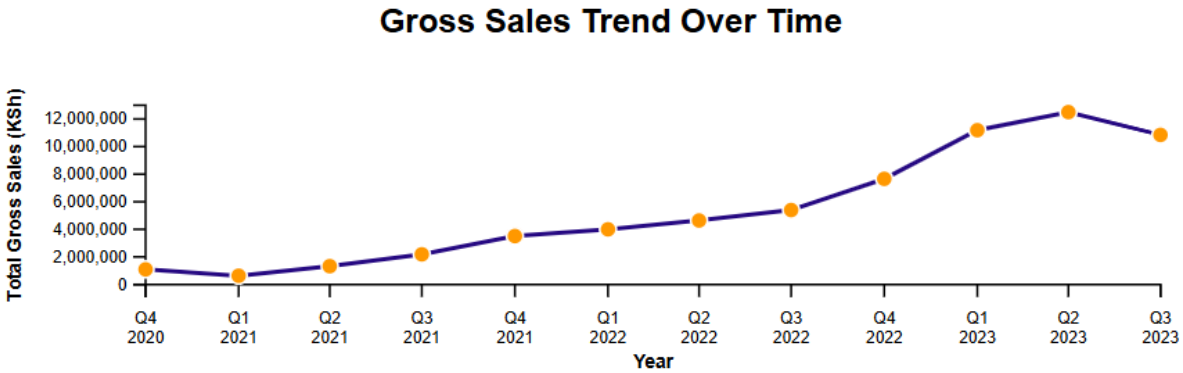
Chart 1: KPI Summary



Purpose: To give a quick snapshot of overall business performance.

Insights: Total sales of KSh 64.98M with over 10K orders. Return rate of 1.55% (169 returns) indicates strong customer satisfaction and product quality. Average order value (Total Sales – Number of Order) of approximately KSh 5975 suggests mid-to-high ticket purchases.

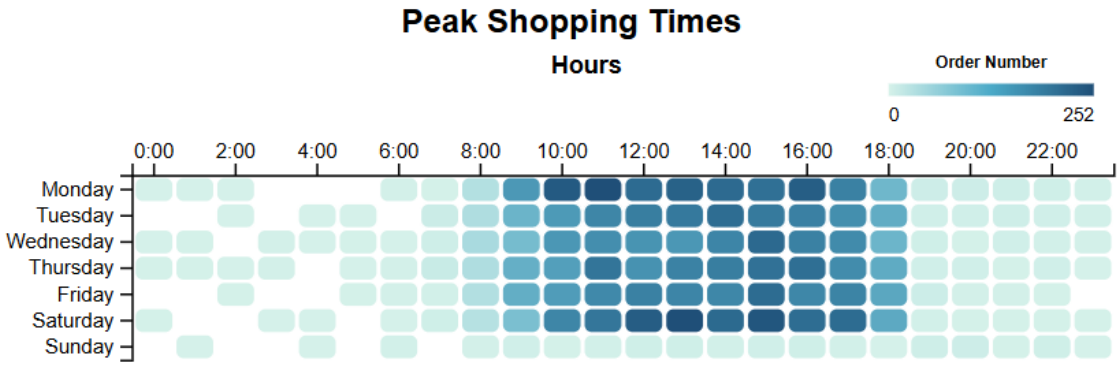
Chart 2: Gross Sales Trend Over Time



Purpose: To visualize how gross sales have changed over time, help businesses to track sales performance and identify seasonal trends, growth or decline periods.

Insights: Sales steadily increased from Q1 2021 to Q2 2023, with a significant increase visible from Q3 2022 onwards, with quarterly sales exceeding KSh 10M by Q1 2023. Slight decline in Q3 2023 may indicate seasonal variation or market saturation requiring strategic intervention.

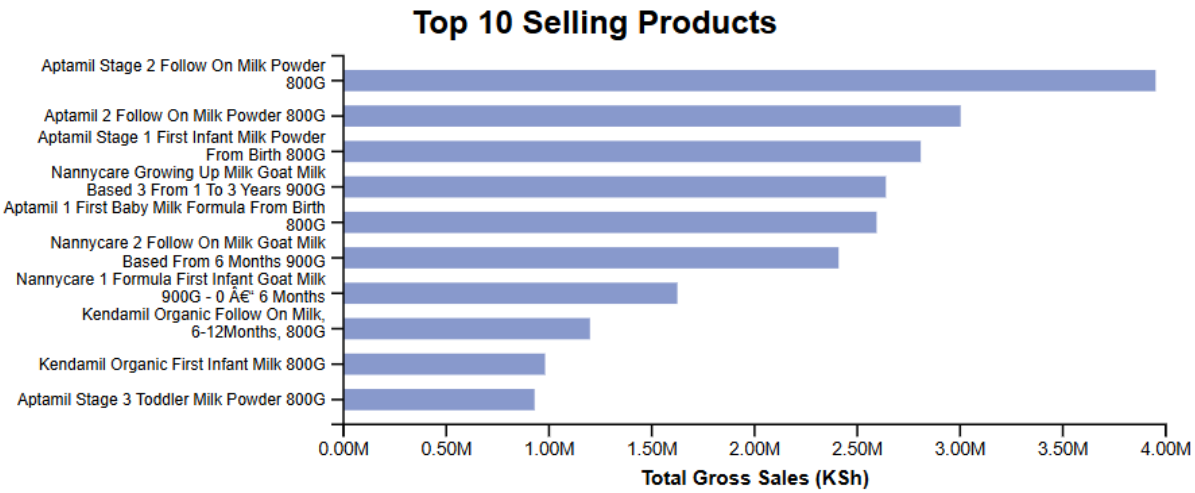
Chart 3: Peak Shopping Times



Purpose: To identify customer shopping behavior patterns by visualizing order frequency across different days and hours, and to set promotion accordingly.

Insight: Order activity concentrated between **9:00 AM and 6:00 PM**, from Monday to Saturday, with the highest order volumes observed during these hours. Sundays show significantly lower engagement, indicating reduced shopping behavior on that day.

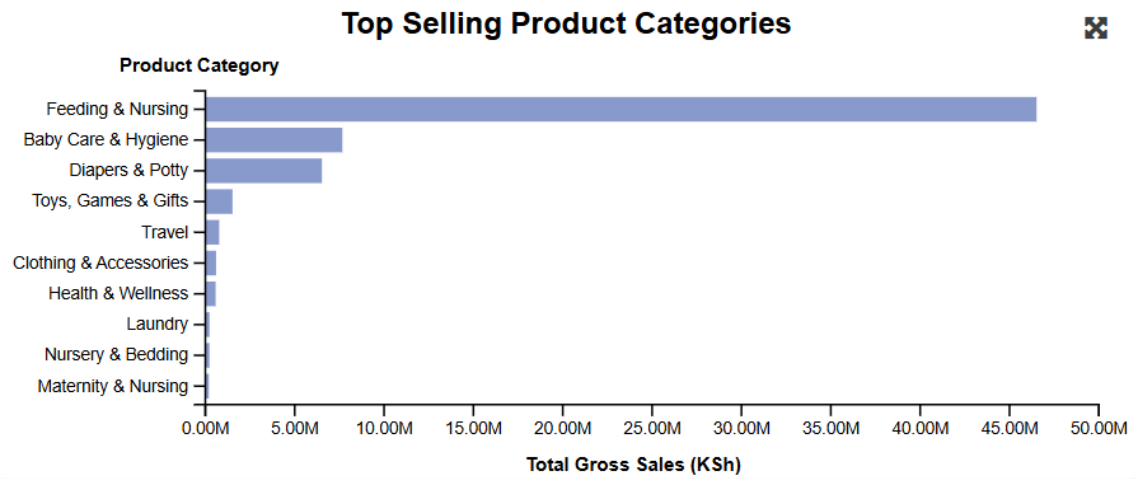
Chart 4: Top 10 Selling Products



Purpose: To highlight the top 10 best-selling products, helping businesses identify high-performing items that drive sales. By focusing on these products, it enables more effective inventory planning and targeted promotional strategies.

Insights: **Baby formula products** dominate the list, with "Aptamil Stage 2 Follow On Milk Powder 800G" leading in sales. This trend indicates strong consumer demand in this category and suggests areas to reinforce stock and marketing efforts.

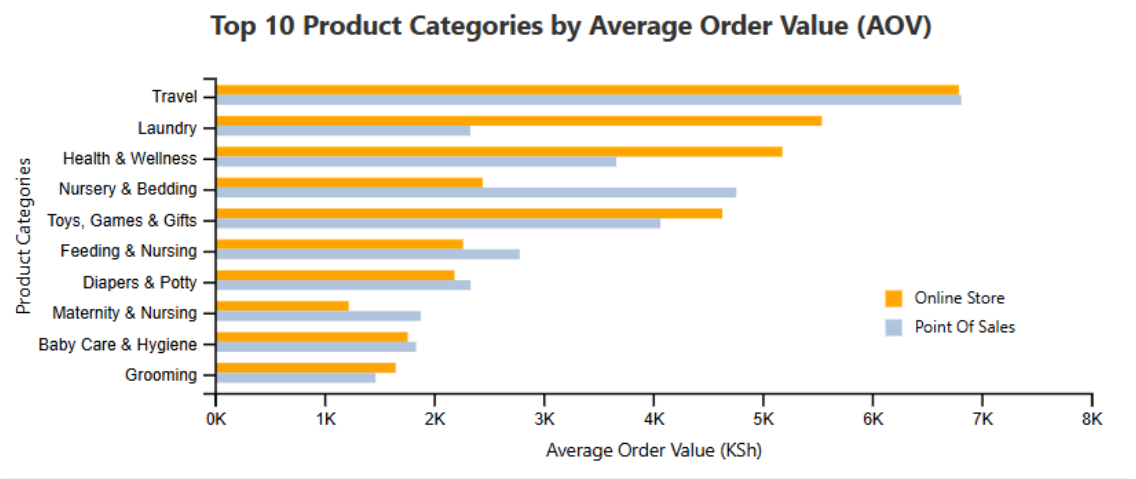
Chart 5: Top Selling Product Categories



Purpose: To highlight which product categories are generating the highest revenue overall, helping business to understand which segments contribute most to total sales. This supports the strategic focus on these categories for marketing prioritization, and product development.

Insights: **Feeding and Nursing** stands out clearly as the most dominant category in sales performance, followed by Baby Care and Hygiene and Diapers and Potty.

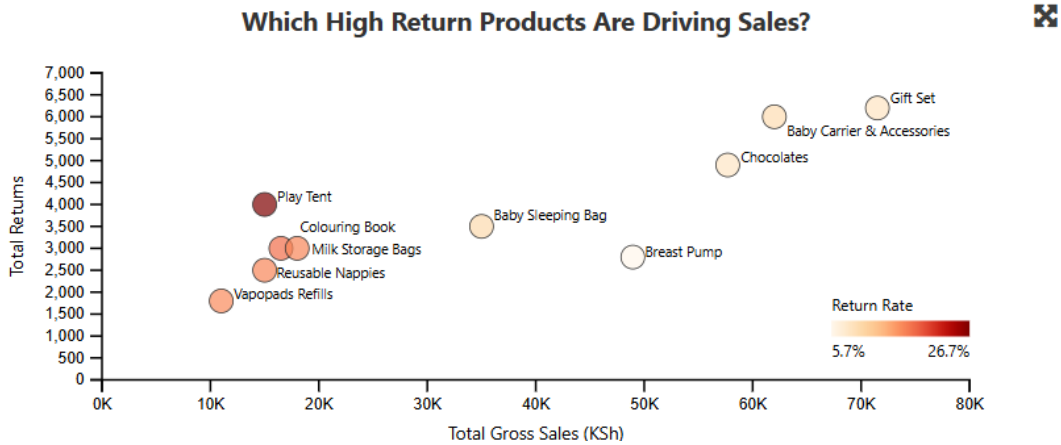
Chart 6: Top 10 Product Categories by Average Order Value (AOV) across Channels



Purpose: To compare the AOV of product categories across online and point of sale channels, help in assessing which categories generate higher revenue per transaction. This enables business to identify premium categories and guides pricing strategy, product bundling, and promotion targeting across sales channels.

Insights: Travel, Laundry, and Health and Wellness lead in AOV, with Travel showing the highest value across both channels. Online store values are generally higher than POS, suggesting **digital buyers tend to spend more per transaction.**

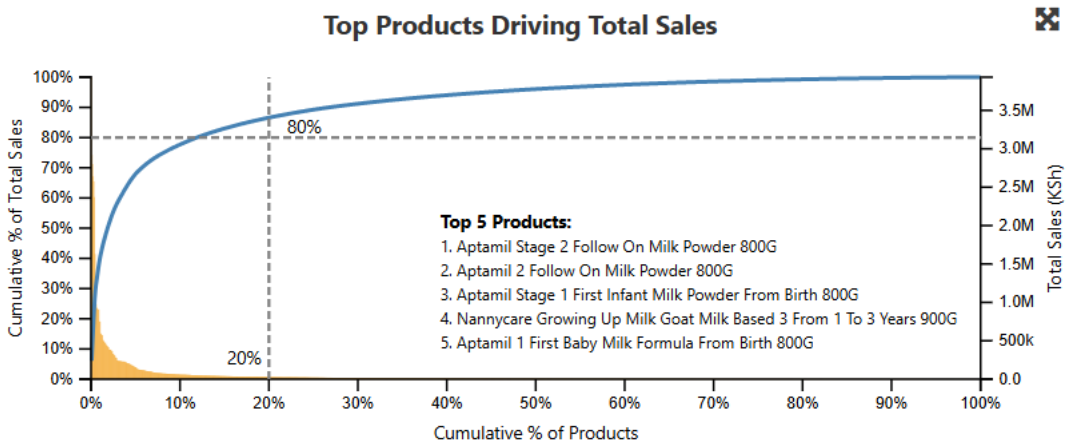
Chart 7: Which High Return Products Are Driving Sales?



Purpose: To identify products that have both high return volumes and sales, helping businesses to understand which items may be affecting revenue and customer satisfaction. This insight supports reviewing product quality, setting clearer expectations, and refining return policies.

Insights: Products like Gift Set and Baby Carrier & Accessories are driving strong sales despite high return volumes, as their return rates are low, indicating strong overall performance. Items such as Play Tent and Colouring Book have high return rates and volume despite lower sales, which may signal quality concerns or customer dissatisfaction.

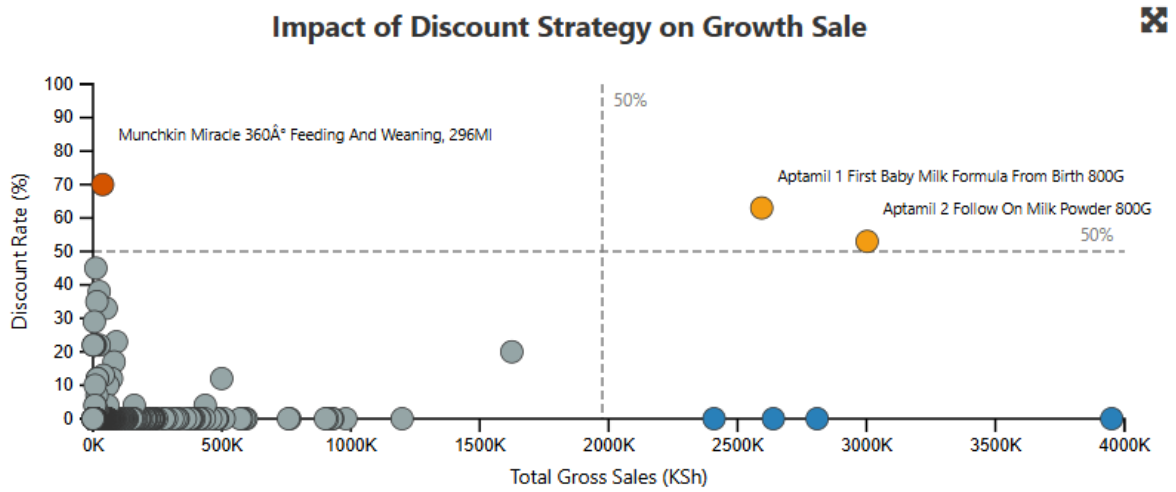
Chart 8: Top Products Driving Total Sales



Purpose: To analyze product contribution to total sales and identify the key contributors that generate the majority of revenue, supporting strategic product prioritization and inventory planning.

Insights: Classic 80/20 rule applies: 20% of products generate 80% of total sales. Top 5 products are all infant/baby milk formulas (Aptamil and Nannycare brands), indicating strong market concentration. High product concentration presents both opportunity (focus marketing) and risk (over-reliance on single category).

Chart 9: Impact of Discount Strategy on Growth Sale



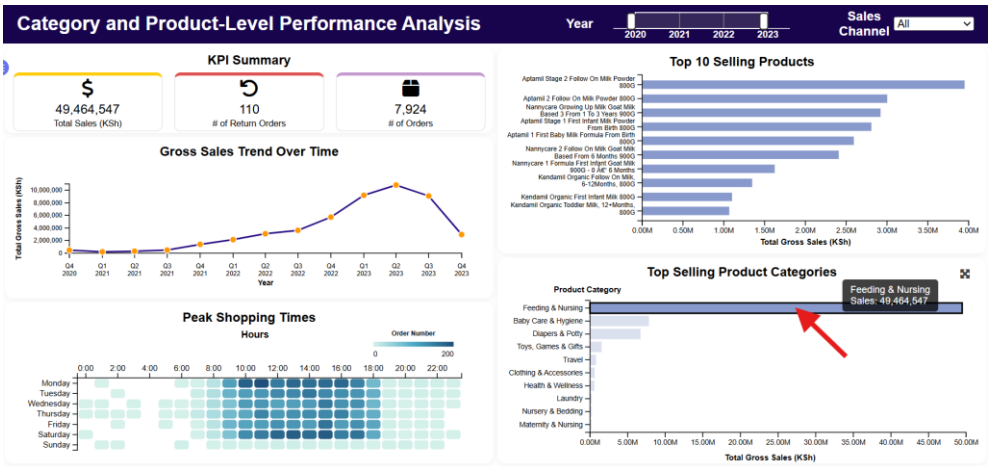
Purpose: To evaluate how different discount strategies affect total gross sales, helping businesses understand the effectiveness of discount approaches in driving revenue. These avoid unnecessary margin loss and focus promotional efforts where they yield the best results.

Insights: Majority products that fall under the Low Discount and High Sales group indicate that high discounts are not always necessary to drive revenue. A few products in light orange dot achieve high sales with high discounts, indicating an effective discount strategy. While products in red dot experiences high discounts but low sales, indicating potential over-discounting with poor returns.

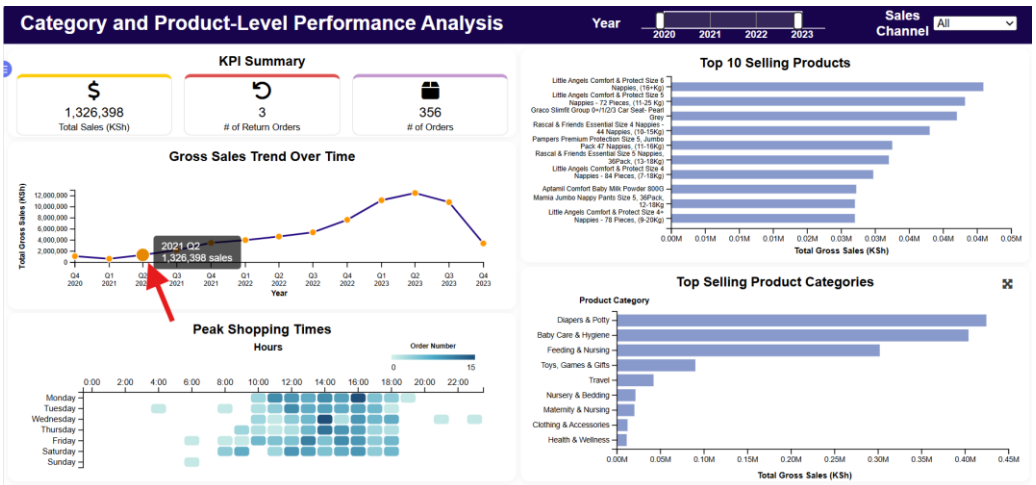
6.0 Interactivity Features

To effectively transform the dataset into actionable insights, the following interactivity features are implemented:

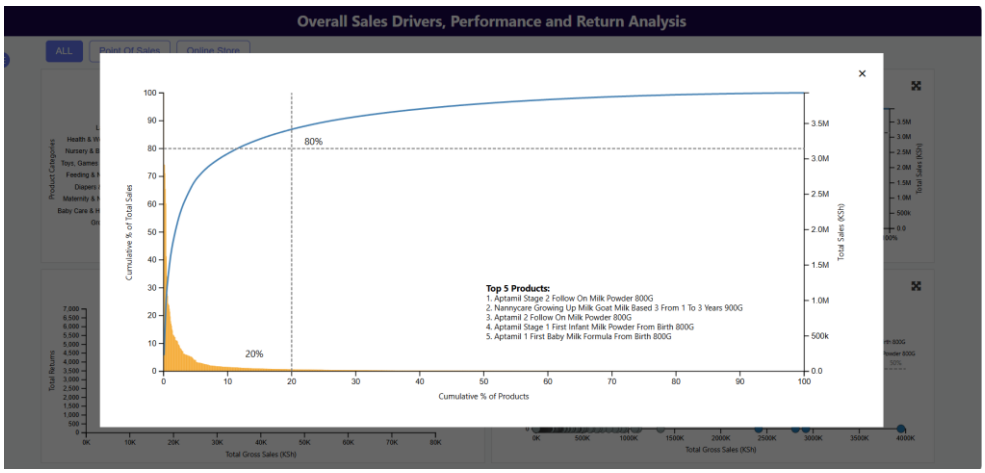
- **Hover Tooltips:** When user hovers over a data point or bar, detailed information such as product name, sales amount, returns, or time is shown. This helps users to better understand the data precisely. The hovered object is also outlined with a thicker border to visually highlight it.
- **Dynamic Linking Across Charts (Filter by product category chart):** When users click a product category bar in the chart, related charts such as top-selling products and peak shopping times update automatically to display information corresponding to that product category.



- **Rollup Interactivity by Quarter of Each Year:** When the users click a point in the line chart representing a specific quarter, other related charts such as top-selling products, product categories, and peak shopping times automatically update to reflect data for that selected quarter. This enables users to explore seasonal patterns and focus on time-specific performance insights.

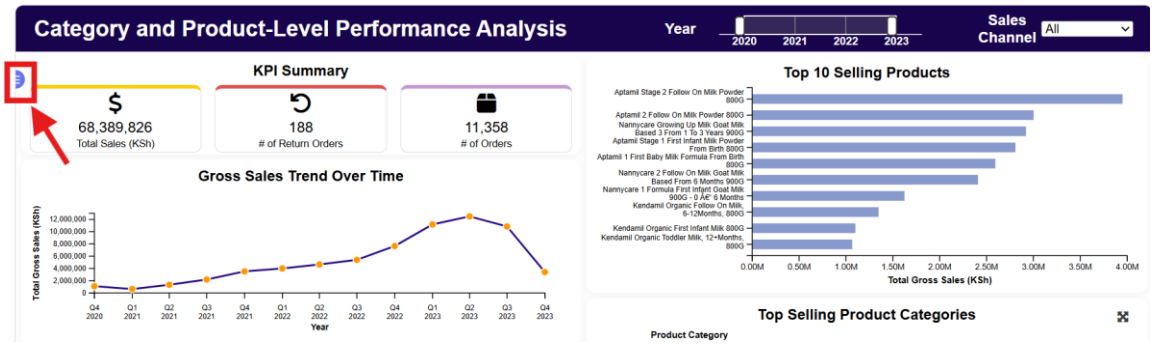


- **Zooming Charts:** When the user clicks the Zoom icon button, the selected chart dynamically enlarges and is displayed in a pop-up window. This allows users to closely examine detailed data points and explore the chart more thoroughly, which enhance insight discovery and data interpretation.



- **Linked Dashboards via Sidebar Navigation**: Two dashboards are interconnected and accessible through sidebar navigation. Users can seamlessly switch between dashboards. This supports a cohesive analytical workflow and broader exploration of the dataset across different dimensions.

Step 1: Click on sidebar icon.



Step 2: Navigate between two dashboards.

7.0 Design Principles - Shaffer's 4Cs

The visualization design principle used is Shaffer's 4Cs.

1. Clear

The dashboard is designed to deliver clear and direct insights. Each visualization addresses a specific business question, such as identifying best-selling products, assessing the sales impact of returns or evaluating effectiveness of discount strategy. KPI cards used clear icons and bold figures to highlight key metrics like total sales, returns, and order volume. This allows audience like analyst and decision makers to understand performance indicators immediately. Charts such as line graph and Pareto chart are labeled clearly with descriptive titles and axis names.

2. Clean

The visualization maintains a clean and consistent design with uniform fonts, colors, and spacing that reduce distraction and enable users to interpret easily. Labels are clear without being overwhelming, and axis titles are clearly defined with appropriate scales. The chart types used are appropriate, for example, heatmap to show peak shopping times, scatter plot to explore the relationship between discounts and sales, and bar charts to compare product or category performance. Interactive features such as tooltips and filters are integrated. This enhances user engagement while keeping the overall layout neat and organized.

3. Concise

The visualizations are rich in insight and remain uncluttered. Each chart is designed to highlight a specific insight without unnecessary detail. For example, the Pareto chart clearly shows that 20% of products contribute to 80% of sales. Besides, AOV comparisons across sales channels are efficiently illustrated with dual-colored bars, summarizing patterns without verbose text. The scatter plot on high-return products also concisely presents gross sales, total returns, and return rates in one clear view, enable viewer to spot revenue-impacting items immediately.

4. Captivating

The dashboard draws attention through effective visual storytelling. The use of colour such as blue for high-performing low-discount products and contrast colour like orange and red for discount strategy analysis makes charts visually engaging. The interactivity features such as hover effects with bold outlines, dynamic filtering by category or quarter, and zoom-in view keeps users engaged and encourage to explore further.